

Making Room for Agriculture

Abstract

This paper explores the balance between agricultural productivity and environmental sustainability through mathematical modeling and optimization frameworks. It addresses key challenges in agricultural ecosystem management by integrating natural processes, human decisions, and optimization strategies.

The natural processes are modeled using three dynamical systems: (1) a **Reaction-Diffusion Model** to simulate the spatial-temporal dynamics of crops and weeds under periodic interventions, (2) a **Tri-Trophic Dynamics Model** to analyze the interactions among crops, pests, and beneficial insects, and (3) a **Tri-Trophic and Soil Vitality Dynamics Model**, incorporating soil nutrients and their feedback effects on ecosystem stability. Specifically, we employ a **Beta (Cardinal) Function** for Temperature and Rainfall to capture the nonlinear suitability of environmental factors for soil and plant health, improving model accuracy.

In analyzing human decisions, we explore scenarios such as **herbicide removal** and **bat integration**. The **Energy-Flow Model**, presented in a logarithmic scale, reveals the cascading effects of species interactions across trophic levels. Additionally, we quantify population dynamics, incorporating fluctuations and confidence intervals to capture variability and uncertainties in ecosystem behavior.

To optimize management strategies, we define two objective functions: (1) the **Economic Objective Function**, which maximizes harvest revenue while minimizing pesticide costs, and (2) the **Environmental Objective Function**, which combines the **SFI** (Soil Fertility Index)—a measure of soil health based on chemical inputs and physical properties, and the **Shannon Index**, which quantifies biodiversity based on species richness and evenness. These functions are normalized to ensure comparability, enabling integrated evaluation of trade-offs between economic and ecological goals. Using optimization techniques, including **SLSQP** (Sequential Least Squares Programming) and **NSGA-II** (Non-dominated Sorting Genetic Algorithm II), we identify Pareto-optimal solutions that balance the conflicting objectives of economic benefits and ecological sustainability. The **Pareto Front** illustrates trade-offs, offering decision-makers a spectrum of optimal strategies that prioritize long-term ecosystem resilience and economic stability.

Our findings highlight the potential of dynamic approaches to sustainable agriculture, providing actionable insights into reducing chemical dependency, enhancing biodiversity, and improving soil health while maintaining economic viability. These strategies pave the way for sustainable agricultural ecosystems resilient to environmental and economic pressures.

Keywords: Dynamics Model, Ecosystem Stability, Objective Functions, NSGA-II

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1 Introduction

1.1 Background

The expansion of agricultural land has been the primary driver of global deforestation, resulting in profound ecological changes over millennia. Historically, forests, wild grasslands, and shrubs once dominated 71% of the Earth's land surface, providing vital ecosystem services such as carbon sequestration, soil stabilization, and biodiversity conservation. However, as agricultural practices expanded, these ecosystems were gradually replaced by croplands and grazing areas. As Figure 1 [1] demonstrates, 46% of the land that once sustained forests and grasslands has been converted to agriculture nowadays.

In the transition from forest to farm, the species that once thrived in the forest face a dramatic shift in their habitat. Many of the original forest species, such as certain birds, insects, and small mammals, may struggle to adapt to the altered landscape. Some might migrate to nearby undisturbed areas, while others may not survive the environmental changes brought about by intensive farming practices. However, over time, the agricultural ecosystem can evolve to support a new set of species. Bats, for example, may take advantage of the pest-rich environment, providing natural pest control. Birds might return to feast on the insects that flourish in the crops. Other species, like pollinators and beneficial insects, can also make their way into the system, contributing to pollination and soil health.

This transformation replaces diverse ecosystems with monoculture farming, diminishing ecological resilience and increasing susceptibility to pests and diseases. To sustain yields, farmers often rely on herbicides and pesticides, which further degrade soil health and even harm some beneficial species. Addressing these challenges requires balancing the trade-offs of modern farming practices by adopting sustainable approaches that restore ecosystem stability and promote long-term agricultural productivity.

1.2 Our Work

Our work addresses the balance between agricultural productivity and environmental sustainability through modeling natural processes, human decisions, and optimization frameworks. As

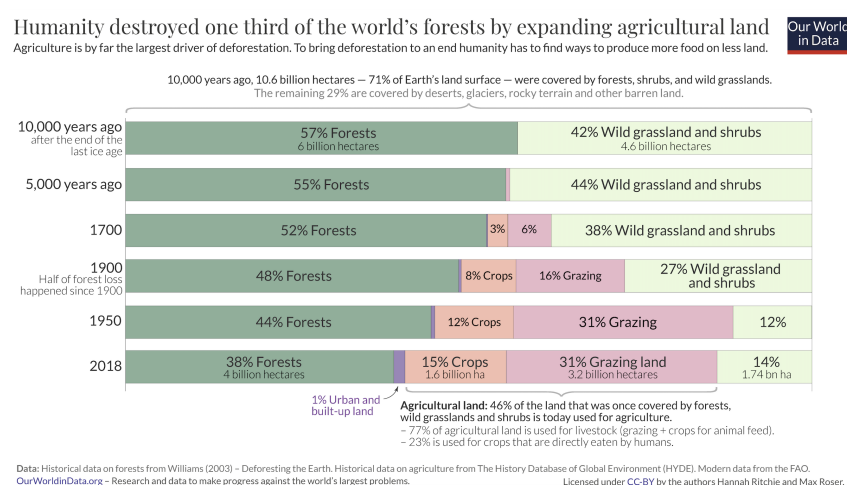


Figure 1: Agriculture: largest driver of Deforestation. Adapted from <https://ourworldindata.org/world-lost-one-third-forests>

Figure 2 illustrates, key models developed include reaction-diffusion dynamics between crops and weeds, tri-trophic dynamics among crops, pests, and beneficial insects, and soil vitality interactions with environmental factors. Human decisions like herbicide removal and the role of bats in pest control are also explored. Two competing objectives—economic (maximizing crop revenue) and environmental (enhancing soil fertility and biodiversity)—are optimized using SLSQP and NSGA-II methods. The results offer actionable recommendations for farmers to optimize pesticide use, harvest schedules, biodiversity, and soil health for long-term sustainability.

2 Assumptions and Notations

2.1 Assumptions

2.1.1 General Assumptions

1. **Environmental Homogeneity:** The farmland domain D is uniform in terms of soil vitality, light availability, and climate conditions.
2. **Interaction Representation:** Interspecies competition, herbicide effects, and harvesting rates are modeled as first-order effects, ignoring higher-order ecological feedbacks.
3. **Population Dynamics:**
 - Most populations exhibit logistic growth [8], with density-dependent constraints such as resource availability and carrying capacities.
 - Carrying capacities (K) are either constant or dynamically dependent on interactions and environmental factors.
4. **Periodic Interventions:** Herbicide, pesticide application and harvesting are periodic and follow specific patterns to simulate regular, predictable management.
5. **Simplified Food Web Dynamics:**
 - Predator-prey and competitive interactions are linear or bilinear, ignoring adaptive behaviors and indirect interactions.

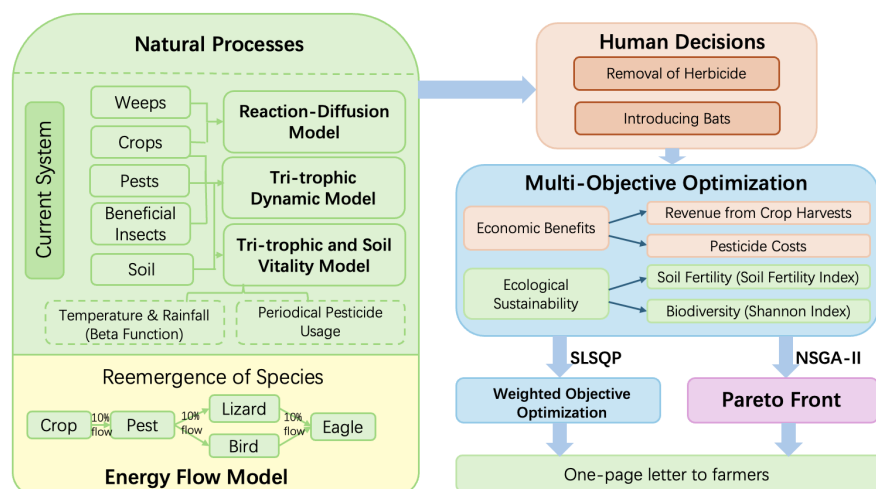


Figure 2: Flow Chart of our Model

- Beneficial organisms and predators do not exhibit feeding preferences.

2.1.2 Assumptions for New Species Introduction & Bats

1. **Ecological Adaptability:** The newly introduced species can adapt to the environmental conditions of the agricultural ecosystem, such as temperature, humidity, and soil properties, ensuring population sustainability.
2. **Introduction Rate:** The introduction of new species occurs gradually rather than at a large scale or rapid pace, allowing the ecosystem sufficient time to adjust.
3. **Population Dynamics:** The population of the new species is constrained by available resources (e.g., food, habitat) and follows a logistic growth model.
4. **Pest Control by Bats:** Bats primarily feed on agricultural pests, and their predation rate is proportional to the pest population density, effectively reducing pest numbers in high-density environments.
5. **Pollination Services by Bats:** Bats provide pollination services to certain crops (e.g., fruits and nuts) during their nocturnal activity, with pollination efficiency depending on the bat population density and crop type.

2.2 Notations

The following variables and equations are used to model the system:

Notations	Description	Units
$P(t)$	Population density of crops	plants/m ²
$Z(t)$	Population density of weeds	plants/m ²
$H(t)$	Population density of pests	insects/m ²
$B(t)$	Population density of beneficial insects	insects/m ²
$N(t)$	Nutrient availability in the soil	plants/m ²
$L(t)$	Population density of Lizards	lizards/m ²
$E(t)$	Population density of Eagles	eagles/m ²
$A(t)$	Population density of Bats	bats/m ²
$G(t)$	Seasonal harvesting rate	plants/(m ² ·month)
$C(t)$	Concentration of chemicals (pesticide C_p , herbicide C_h)	g/L
$W(t)$	Organic waste content in the soil	kg/m ²
$T(t)$	Temperature	°C
$R(t)$	Erosion rate	kg/(m ² ·month)
$\phi(T(t))$	Effect of temperature on nutrient availability	dimensionless
$\phi(R(t))$	Effect of erosion on nutrient availability	dimensionless
K_P	Carrying capacity of crops	plants/m ²
K_Z	Carrying capacity of weeds	weeds/m ²
K_H	Carrying capacity of pests	insects/m ²
K_B	Carrying capacity of beneficial insects	insects/m ²
r_1, r_2, r_3	Growth rates for crops, pests, and beneficial insects	1/month
w_1, w_2, w_3	Interaction coefficients between populations	1/(insects·m ²)

3 Natural Processes

3.1 The Current System

3.1.1 Reaction-Diffusion Model of Crops and Weeds

We first consider a coupled reaction-diffusion model to simulate the spatial and temporal dynamics of crops $P(t)$ and weeds $Z(t)$ in an agricultural ecosystem. Suppose our domain D is the total area of the farmland. The model incorporates key ecological processes, including logistic growth, interspecies competition, and diffusion, to capture the interactions between weeds and crops. Periodic herbicide application, modeled as $h_Z = h_{Z0} \cdot (1 + \sin(\omega_Z t))$, is introduced to control weed proliferation. Similarly, the crop harvesting rate, modeled as $h_P = h_{P0} \cdot (1 + \sin(\omega_P t))$, influences crop density dynamics. Neumann boundary conditions (zero flux) are applied to simulate a closed system, and finite difference methods are used to approximate the diffusion terms. The PDE system is:

$$\frac{\partial Z}{\partial t} = r_Z Z \left(1 - \frac{Z + \alpha P}{K_Z} \right) + D_Z \nabla^2 Z - h_Z Z$$

$$\frac{\partial P}{\partial t} = r_P P \left(1 - \frac{P + \beta Z}{K_P} \right) + D_P \nabla^2 P - h_P P$$

$$\frac{\partial Z}{\partial n} = 0 \quad \text{and} \quad \frac{\partial P}{\partial n} = 0 \quad \text{on } \partial D \times [0, T],$$

Specifically, the parameters are the growth rates r_Z and r_P , carrying capacities K_Z and K_P , and competition coefficients α and β . The spatial spread of weeds and crops is modeled using the Laplacian operator ∇^2 , with diffusion coefficients D_Z and D_P .

Figure 3 and 4 illustrate the spatial and temporal dynamics of weeds and crops. Initially, weeds exhibit higher densities (initial $Z \sim 20 - 30$) and broader spatial distribution due to their higher growth rate ($r_Z = 0.8$) and carrying capacity ($K_Z = 120$), and diffusion coefficient ($D_Z = 0.1$), while crops show lower densities (initial $P \sim 5 - 15$) and more localized distribution due to lower growth rate ($r_P = 0.4$) and diffusion coefficient ($D_P = 0.05$). The oscillatory patterns in both densities reflect the influence of periodic interventions, such as herbicide application (h_Z)

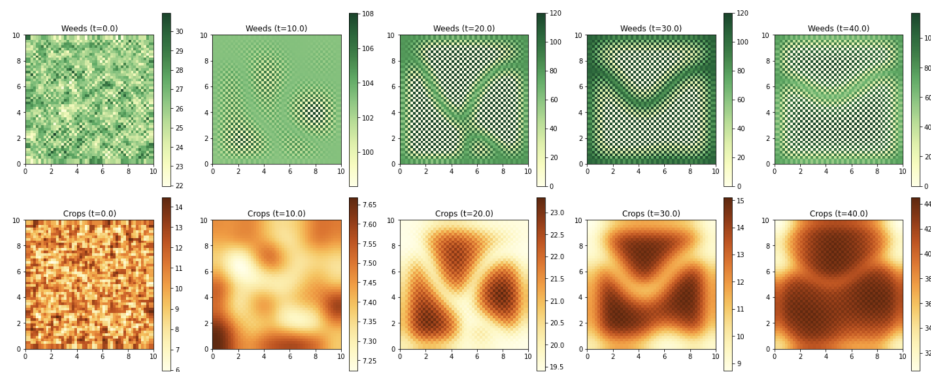


Figure 3: Spatial Distribution of Weeds and Crops Over Time

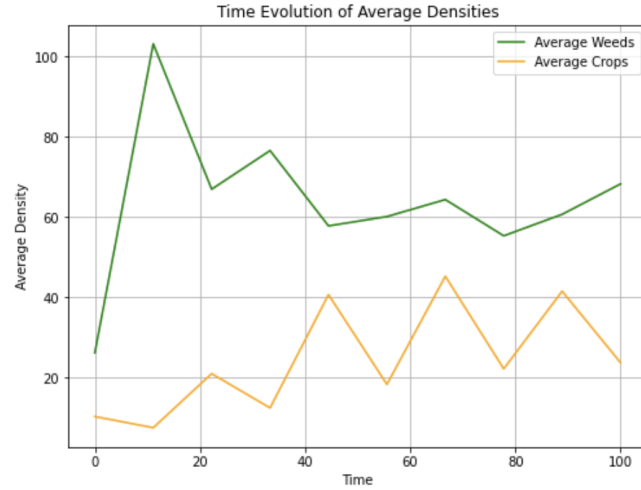


Figure 4: Time Evolution of Average Weed and Crop Densities

for weeds, with $h_{P0} = 0.15$ and $\omega_P = 0.3$, and harvesting for crops (h_P), with $h_{P0} = 0.15$ and $\omega_P = 0.3$. Competition coefficients ($\alpha = 0.7$ and $\beta = 0.5$) further modulate the interactions.

Unlike the one-time impact of harvesting, herbicide application provides sustained weed control over an extended period. Therefore, instead of merely applying the sine function, we adopt a more comprehensive model [5] which captures additional factors affecting h_Z and its prolonged effects, which is expressed as,

$$C_h(t) = h_z(t) = f(\text{decay rate, assimilation rate, dosage amount, } \dots)$$

Our objective is to determine the optimal dosage amounts and intervals such that the chemical concentration reaches the lowest effective level L as quickly as possible and thereafter stays within the range $[L, H]$, where H is the highest safe level. We denote C_{i-1} as the concentration at the beginning of the i -th interval, R_i as the residual concentration at the end of the i -th interval.

The residual concentration R_n at the end of n intervals can be expressed as:

$$R_n = C_0 e^{-kT} + C_0 e^{-2kT} + \dots + C_0 e^{-nkT} = \frac{C_0 e^{-kT} (1 - e^{-nkT})}{1 - e^{-kT}}$$

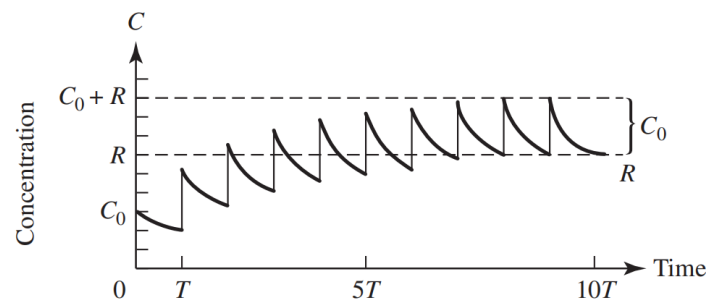


Figure 5: $C(t)$ with appropriate dosage amount and interval

As the number of intervals $n \rightarrow \infty$, the residual concentration R approaches a steady-state value:

$$R = \lim_{n \rightarrow \infty} R_n = \frac{C_0 e^{-kT}}{1 - e^{-kT}}$$

If we set the initial dosage amount C_0 as $H - L$, the interval T as $\frac{1}{k} \ln \frac{H}{L}$, then as Figure 5 indicates, the concentration of the herbicide always remains between the lowest effective level and highest safe level, thus delivering long-lasting effects.

Figure 6 and 7 illustrate the spatial and temporal dynamics of weeds and crops with improved model of herbicide concentration. Initially, the herbicide concentration had not yet reached the minimum effective level, so its suppressive effect on weeds had not yet begun. During this phase, both weeds and crops experienced growth due to sufficient space and resources. Around $t = 17$, the herbicide likely reached the minimum effective concentration, triggering a noticeable decline in weed density. At the same time, as the competition for resources decreased, crop density began to rise. Following this turning point, the herbicide's continued effect, along with increasing competition between the two species, led to a gradual decline in both weed and crop densities. Over time, the system approached stability, with both densities leveling off, indicating a balance maintained by the periodic herbicide application and interspecies competition. This dynamic demonstrates the effectiveness of the improved herbicide model in suppressing weeds while promoting crops under competitive constraints.

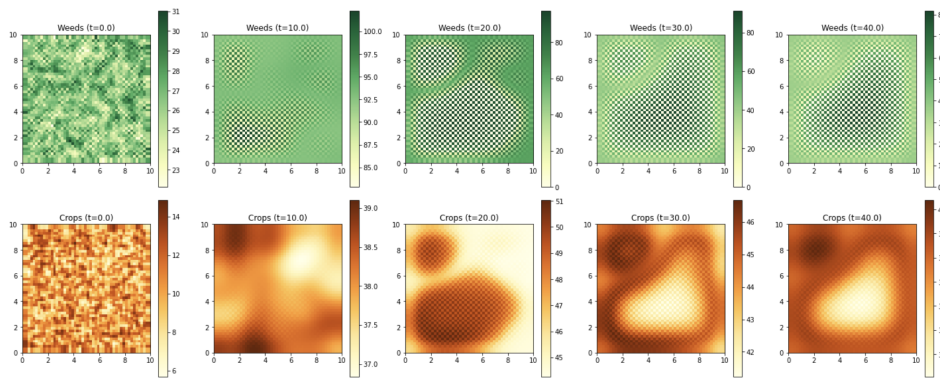


Figure 6: Spatial Distribution of Weeds and Crops For Improved h_Z

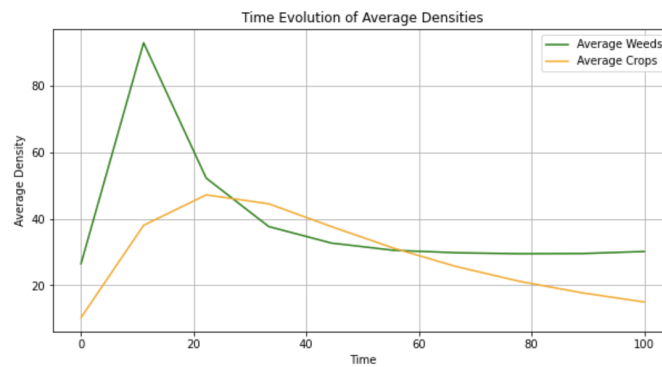


Figure 7: Time Evolution of Average Weed and Crop Densities For Improved h_Z

3.1.2 TriTrophic Dynamics Model

In the next model, we are examining the interaction between different components of an agricultural ecosystem that involves crops (Producer), pests (Herbivores), and beneficial insects (Carnivores), which have a dynamic relationship affecting agricultural stability. The system also considers the influence of human interventions such as chemical use, fertilizer application, and organic waste management on the ecosystem. The variables and parameters in the system govern the interactions among the species and the environmental factors that affect them.

We now proceed to the following system of ordinary differential equations that describe the evolution of these populations over time:

$$\begin{aligned}\frac{dP}{dt} &= \frac{r_1}{1+C} P \left(1 - \frac{P+H}{K_P} \right) - \alpha_I I_0 P, \\ \frac{dH}{dt} &= \beta_H H \left(1 - \frac{H-P+B}{K_H} \right) - C_P H, \\ \frac{dB}{dt} &= \beta_B B \left(1 - \frac{B-H}{K_B} \right) - \mu_B B.\end{aligned}$$

The crop P grows logistically, with a reduced growth rate $\frac{r_1}{1+C}$, limited by competition with pests H and environmental light intensity $-\alpha_I I_0 P$. The population of pests H also exhibits logistic growth ($\beta_H H$), constrained by competition with crop P and beneficial insects B , and subject to concentration of pesticides $-C_P H$. The beneficial insects B grow logistically ($\beta_B B$), limited by competition with pests H , and decays naturally $-\mu_B B$.

The analysis of Figure 8, based on the coupled ODE model, reveals intricate dynamics among crops (P), pests (H), and beneficial insects (B) in an agricultural ecosystem. Figure 8(a) shows an initial sharp decline in crop population due to rapid pest growth, followed by a decrease in pests as beneficial insects increase; over time, all populations stabilize into periodic cycles, with crops remaining low, pests exhibiting moderate oscillations, and beneficial insects showing smoother fluctuations. In Figure 8(b), the system initially exhibits similar behavior as Figure 8(a). However, since (β_H) becomes much smaller, only 0.002, this suppression occurs after a significant decline in the crop population, leading to a temporary collapse where all three populations drop to near-zero levels. Following this collapse, the system resets, and a

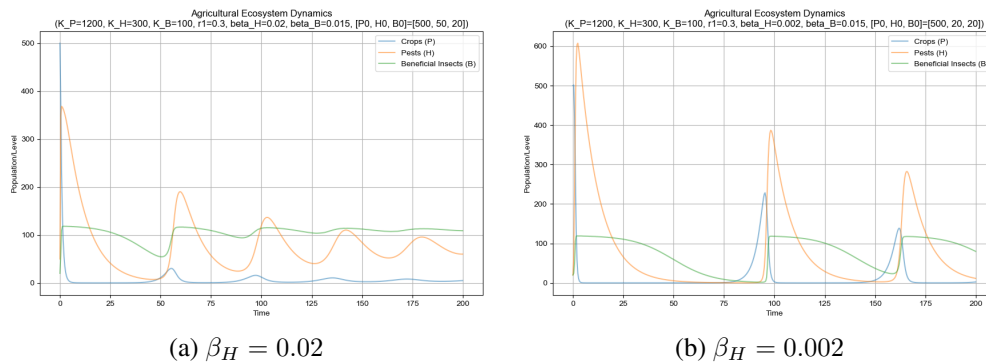


Figure 8: Dynamic Ecosystems for TriTrophic Model with Fixed Capacity

new cycle begins with crops regrowing, pests re-emerging, and beneficial insects rebuilding their population. This cyclical pattern highlights the delicate balance required in agricultural ecosystems, where overpopulation of pests can lead to temporary ecosystem collapse, followed by recovery.

In the tritrophic dynamics model, we can further extend the expression of pesticide concentration $C_p(t)$. Denote $C_p(t)$ with seasonal periodicity defined as:

$$C_{\text{seasonal}}(t) = \left(\sum_{i=1}^n C_0 e^{-k \cdot i \cdot T} \right) \cdot \left(1 - \cos \left(\frac{2\pi t}{\text{period}} \right) \right) \quad (1)$$

where: C_0 is the initial dose of the pesticide, k is the decay rate of the pesticide, T is the interval between pesticide applications, and period is the seasonal period (in weeks, as each time step corresponds to one week). The sinusoidal factor $\left(1 - \cos \left(\frac{2\pi t}{\text{period}} \right) \right)$ ensures higher pesticide concentrations in summer (peaking at $t = \text{period}/2$) and lower concentrations in winter (minimal at $t = 0, \text{period}$).

Figure 9 reflects the impact of seasonal periodicity on pesticide concentration and its cascading effects on crop, pest, and beneficial insect populations. The incorporation of the sinusoidal factor introduces clear annual fluctuations in pesticide concentration, with higher levels during summer and lower levels during winter. This seasonal variation is mirrored in the population dynamics of the ecosystem. Crops benefit from higher pesticide concentrations in summer, as pest populations are significantly reduced during this period. Conversely, in winter, as pesticide levels drop, pest populations recover, leading to a decline in crop populations. Beneficial insects, while also affected by pesticide levels, exhibit more nuanced dynamics due to their dependence on pests as a food source and the indirect effects of changing pesticide concentrations. Over the simulated four-year period, the populations stabilize into a repeating annual cycle, highlighting the interplay between pesticide application and ecosystem stability.

3.1.3 TriTrophic and Soil Vitality Dynamics Model

To address the unstable dynamics in the ecosystem and incorporate the effects of nutrient availability in the soil, we now modify our previous ODE system by controlling the capacity of each species and introducing an additional variable representing soil nutrients:

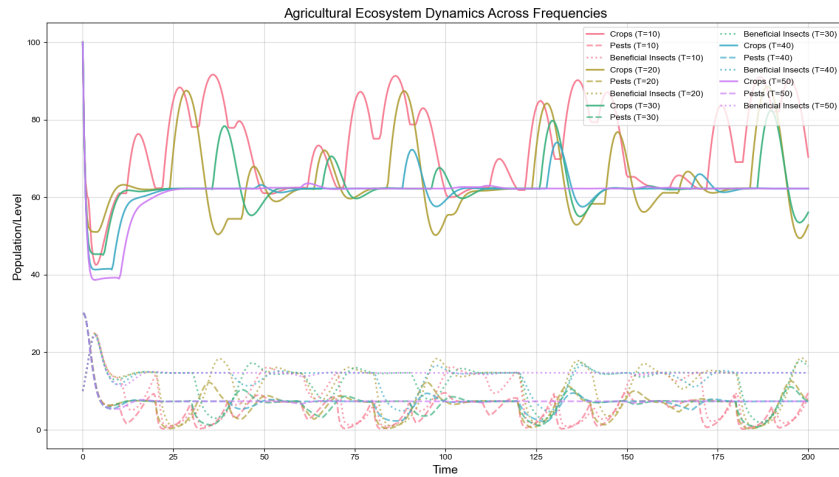


Figure 9: Seasonal Dynamics of Pesticide Concentration and Ecosystem Interactions

1. Crops (Producer):

$$\frac{dP}{dt} = \frac{r_1}{1+C}((N - w_1HP) - GP)$$

This equation models the growth and consumption of crops. Crops grow based on the availability of nutrients $N(t)$, and they are consumed by pests at a rate proportional to the pest population. The seasonal harvesting function $G(t)$ reduces the crop population seasonally. $C(t)$ represents the concentration function of chemicals (pesticide in this case).

2. Pests (Herbivores):

$$\frac{dH}{dt} = r_2H \left(1 - \frac{H}{w_2P} - \beta B \right) - CH$$

This equation describes the growth of the pest population. The pest population increases according to the availability of crops, subject to density-dependent factors (represented by $\frac{H(t)}{w_2P(t)}$) and the predation pressure from beneficial insects $B(t)$.

3. Beneficial Insects (Carnivores):

$$\frac{dB}{dt} = r_3B \left(1 - \frac{B}{w_3H} \right)$$

The beneficial insects, which serve both as predators of pests and consumers of crops, grow according to the availability of pests and their own carrying capacity. The predator-prey dynamics between $B(t)$ and $H(t)$ are modeled with the density-dependent growth term.

4. Nutrients (Soil vitality):

$$\frac{dN}{dt} = r_4(\alpha_w W(t) + f_S(S(T, R)) - \alpha_N N(t))$$

where $N(t)$ represents the soil vitality or soil nutrient index, $\alpha_w W(t)$ is the positive contribution from soil organic matter W , and $f_S(S(T, R))$ represents the direct positive impact from the temperature-rainfall suitability function $S(T, R)$. For example, $f_S(S(T, R))$ could be set as $\alpha_S(S - 1)$. The term $-\alpha_N N(t)$ accounts for self-limitation or decay of soil vitality, and r_4 is the rate parameter for changes in soil vitality.

Soil organic matter equation:

$$\frac{dW}{dt} = I_W(t) - k_W S(\phi(T), \phi(R)) W(t).$$

where $W(t)$ represents the soil organic matter content (e.g., in kg/ha), $I_W(t)$ is the external organic input (e.g., decomposer, manure, crop residues), which may vary seasonally, k_W is the baseline decomposition rate constant, and $S(\phi(T), \phi(R))$ is the suitability factor (from the beta function model), which modulates the rate of decomposition. When $S = 1$, decomposition occurs at its maximum rate $k_W W$, and when $S = 0$, it is minimal. The function $S(\phi(T), \phi(R))$ is defined as follow:

Beta (Cardinal) Function for Temperature and Rainfall

Both temperature $\phi_T(T)$ and rainfall $\phi_R(R)$ may be represented by a beta-type (cardinal) function [7] as shown in Figure 10:

$$\phi_T(T) = \frac{r_T}{T_{\max}} = \left[\left(\frac{T - T_{\min}}{T_{\text{opt}} - T_{\min}} \right) \left(\frac{T_{\max} - T}{T_{\max} - T_{\text{opt}}} \right)^{\frac{T_{\max} - T_{\text{opt}}}{T_{\text{opt}} - T_{\min}}} \right]^c$$

and similarly for $\phi_R(R)$.

$$\phi_R(R) = \frac{r_R}{R_{\max}} = \left[\left(\frac{R - R_{\min}}{R_{\text{opt}} - R_{\min}} \right) \left(\frac{R_{\max} - R}{R_{\max} - R_{\text{opt}}} \right)^{\frac{R_{\max} - R_{\text{opt}}}{R_{\text{opt}} - R_{\min}}} \right]^c$$

Then one can define

$$S(T, R) = \phi_T(T) \times \phi_R(R),$$

giving a two-dimensional suitability measure in the range $[0, 1]$. Whenever temperature or rainfall is far from its respective optimum, $S(T, R)$ becomes low, slowing decomposition and possibly other soil-related processes.

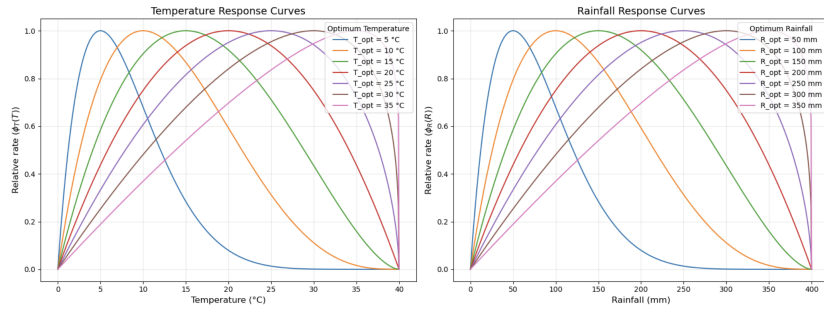


Figure 10: Beta Functions of $\phi_T(T)$ and $\phi_R(R)$

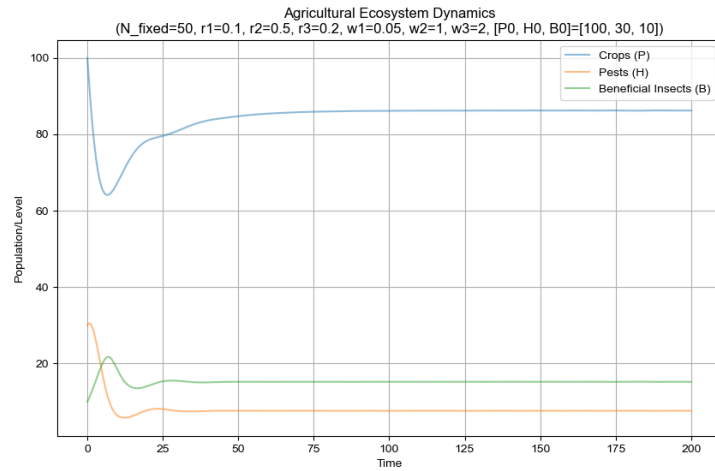


Figure 11: Capacity influenced by Crops with $N_{\text{fixed}} = 50$

In order to simplify the simulation process, we fixed the nutrition from soil vitality as $N = 50$ and set the initial population for the system is $[P_0, H_0, B_0] = [100, 30, 10]$. As Figure 11 demonstrates, under the influence of pests, crop numbers begin to decline as the nutrient supply from the soil is insufficient to offset the damage caused by the pests. However, birds help reduce pest populations through predation. Around $t = 15$, the pest population decreases to its minimum, allowing crops to begin growing again under the influence of soil vitality. Eventually, the system reaches a balance, with the populations of crops, pests, birds, and soil vitality stabilizing in a dynamic equilibrium.

In sum, the system of equations describes the complex interactions between crops, pests, beneficial insects, and the environment. Understanding the dynamics of these populations is crucial for managing agricultural practices and sustaining ecological balance.

3.2 The Reemergence of Species

As time progresses, edge habitats start to mature, leading to the return of native species. With the reintroduction of these species, the agricultural ecosystem undergoes changes influenced by the interactions between these species and the existing environment. The following model considers the interactions between crops ($P(t)$), pests ($H(t)$), beneficial insects ($B(t)$), **lizards** ($L(t)$), and **eagles** ($E(t)$) with dynamically dependent carrying capacities.

1. **Crop population** $P(t)$:

$$\frac{dP}{dt} = \frac{r_1}{1+C}(N - w_1HP) - GP$$

2. **Pest population** $H(t)$:

$$\frac{dH}{dt} = r_2H \left(1 - \frac{H}{K_H}\right) - w_2BH - w_3LH - CH$$

3. **Bird population** $B(t)$:

$$\frac{dB}{dt} = r_3B \left(1 - \frac{B}{K_B}\right) - w_4EB$$

4. **Lizard population** $L(t)$:

$$\frac{dL}{dt} = r_4L \left(1 - \frac{L}{K_L}\right) - w_5EL$$

5. **Eagle population** $E(t)$:

$$\frac{dE}{dt} = r_5E \left(1 - \frac{E}{K_E(B, L)}\right) + w_6(EB + EL)$$

To analyze the system, we could linearize of the system at the equilibrium points.

Let the system be defined as:

$$\frac{d\mathbf{x}}{dt} = \mathbf{F}(\mathbf{x}),$$

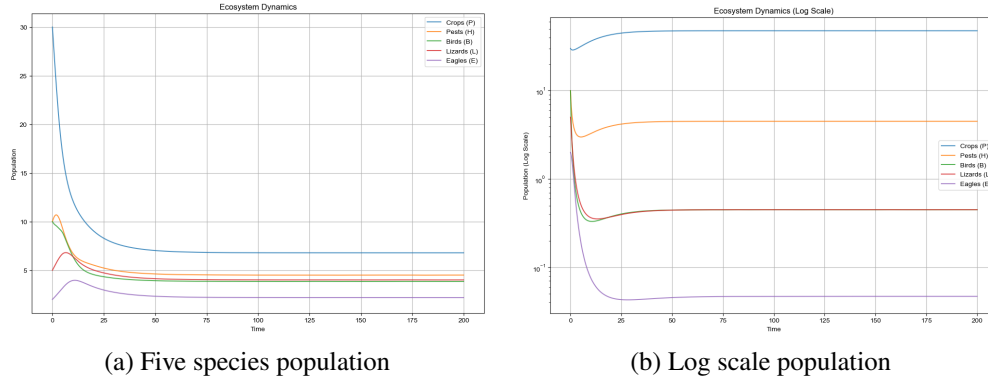


Figure 12: Dynamic Ecosystem After Species Reemerge

where $\mathbf{x} = (P, H, B, L, E)^T$.

Near the equilibrium point $\mathbf{x}^* = (P^*, H^*, B^*, L^*, E^*)$, let

$$\mathbf{x} = \mathbf{x}^* + \boldsymbol{\epsilon},$$

where $\boldsymbol{\epsilon}$ represents a small perturbation. Expanding $\mathbf{F}(\mathbf{x})$ in a Taylor series around $\mathbf{x}^*$ and retaining only the first-order terms gives:

$$\frac{d\boldsymbol{\epsilon}}{dt} = D\mathbf{F}(\mathbf{x}^*)\boldsymbol{\epsilon},$$

where $D\mathbf{F}(\mathbf{x}^*)$ is the Jacobian matrix of $\mathbf{F}(\mathbf{x})$ evaluated at the equilibrium point $\mathbf{x}^*$.

Thus, the linearized system can be expressed as:

$$\frac{d\boldsymbol{\epsilon}}{dt} = J\boldsymbol{\epsilon},$$

After analyzing the equilibrium points in the linearized system, we derived the ecosystem dynamics graph for the five species shown in Figure 12(a). The figure demonstrates that crops decline due to the impact of pests, which rapidly increase in a short period. Following $t = 5$, the pest population begins to decrease quickly. By $t = 12$, the populations of lizards and eagles also start to decline. Eventually, by $t = 50$, the system reaches equilibrium and stabilizes.

In order to improve this ecosystem energy flow model, we set $K_B(H) = 0.1 \cdot H(t)$, meaning the carrying capacity of birds is proportional to 10% of the pest population at any given time. Similarly, the dependent carrying capacities of other trophic levels are set to 10% of the corresponding biomass, reflecting the 10% energy transfer efficiency rule [4].

As the ecosystem reaches equilibrium, the biomass distribution across trophic levels follows a consistent logarithmic scale pattern demonstrating in Figure 12(b). This explains why, when plotted on a logarithmic y-axis, the trophic levels appear evenly spaced. For instance, lizards and birds occupy similar ecological niches in terms of biomass, as their energy sources (pests and birds) align with the proportional energy transfer model. The visualization of Figure 13 underscores the energy constraints in ecosystems and the predictable decline in biomass as energy flows upward through the food web.

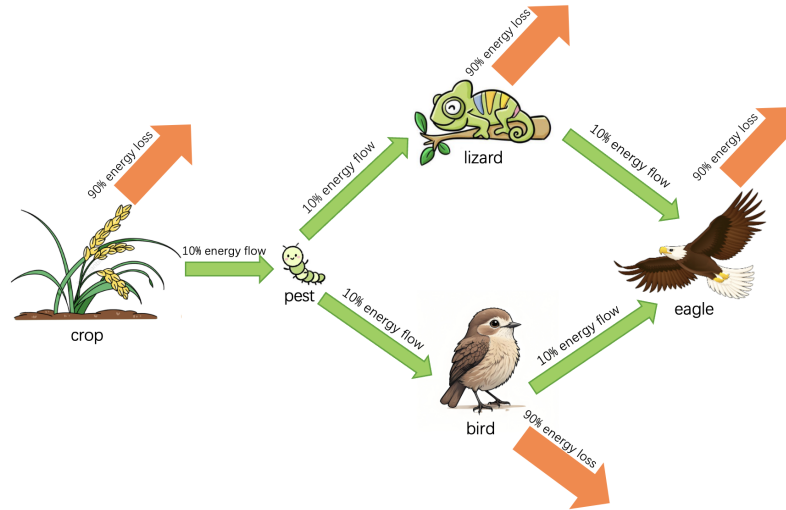


Figure 13: Energy Flow in the Food Web

4 Human Decisions

4.1 Removal of Herbicide

As the ecosystem matures, farmers may attempt to reduce chemical dependency by removing herbicides. This scenario explores the dynamics of an agricultural ecosystem involving crops (P), weeds (Z), pests (H), and beneficial insects (B), under the assumption that humans do not use herbicides ($C_h = 0$). The ODE model for this system is:

$$\frac{dP}{dt} = \frac{r_p}{1 + C_p} P \left(1 - \frac{P + \gamma Z + wH}{K_P} \right),$$

$$\frac{dZ}{dt} = r_z Z \left(1 - \frac{Z + \alpha P}{K_Z} \right),$$

$$\frac{dH}{dt} = r_h H \left(1 - \frac{H + \beta B}{K_H} \right) - C_p H,$$

$$\frac{dB}{dt} = r_b B \left(1 - \frac{B}{K_B} \right).$$

The crop population grows logistically but is hindered by competition from weeds (Z) and pests (H), scaled by competition coefficients γ and w , respectively. Weeds grow logistically and compete with crops, while pests grow logistically and are suppressed by beneficial insects (B), scaled by the competition coefficient β .

Figure 14 indicates that over time, crops decline rapidly to near extinction due to competition with weeds and pest pressure, while weeds dominate the system, stabilizing near their carrying capacity. Pests maintain a higher population than beneficial insects, despite predation, indicating an imbalance in ecological control. Confidence intervals (shaded regions) highlight the sensitivity of crops and weeds to environmental fluctuations, while pests and beneficial insects

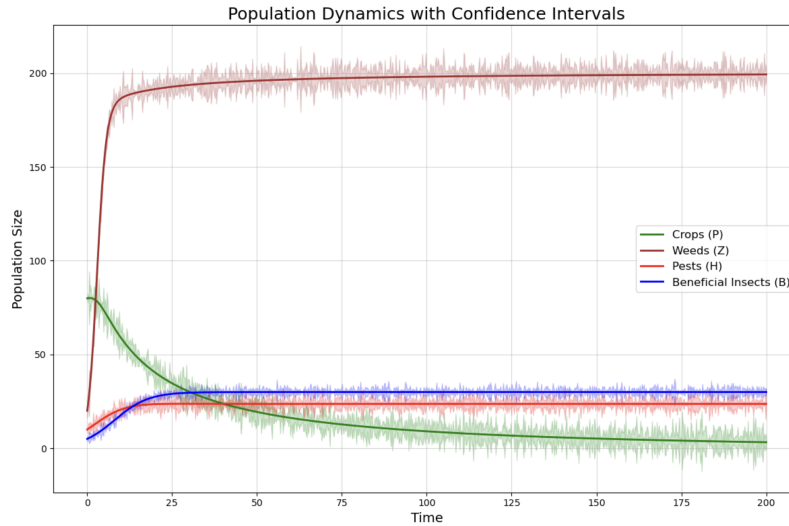


Figure 14: Population Dynamics with Confidence Interval

show more stable dynamics. The system ultimately reaches a dynamic equilibrium where weeds dominate, crops vanish, and both pests and beneficial insects persist.

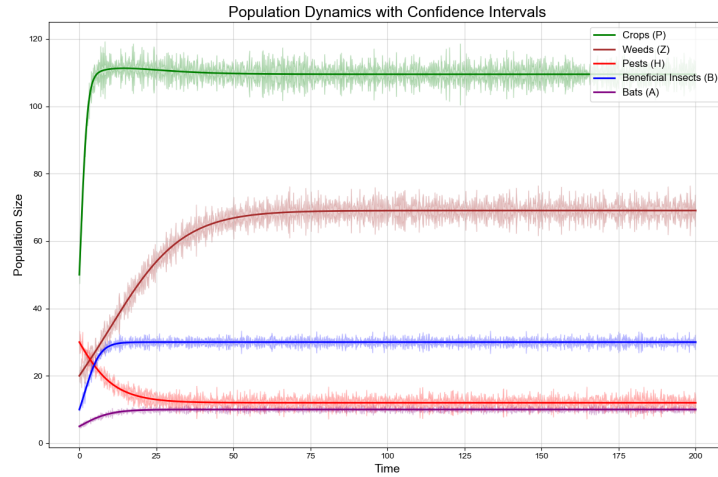
4.2 The Influence of Bat

From discussion in 3.1.1, we find that removal of herbicide will result in an imbalance in ecological control. Incorporating biological controls, such as bats, offers a sustainable alternative to restore stability. Bats play a dual role in ecosystems: as insectivores, they control pest populations, reducing pressure on crops, and as pollinators, they promote plant reproduction and diversity.

This study extends an ordinary differential equation model to incorporate bats into the agricultural food web, highlighting their interactions with crops $P(t)$, weeds $Z(t)$, pests $H(t)$, beneficial insects $B(t)$, and Bats $A(t)$.

In this model, bats (A) and crops (P) exhibit a mutually beneficial relationship: Bats positively impact crops by enhancing growth through pollination, represented by the term $\frac{\delta_p A}{1+kA}$. As the bat population (A) increases, the crop growth rate (r_p) improves, although this effect saturates at higher bat populations where pollination efficiency stabilizes. Conversely, crops support bats by providing essential resources, such as nectar, represented by the term $\delta_a P A$. An increase in the crop population (P) boosts the growth rate of the bat population, aiding in its reproduction and sustainability.

This bidirectional relationship demonstrates mutualism: bats promote crop reproduction through pollination, while crops provide essential resources to sustain and stabilize the bat population. The model evaluates the effects of bats and alternative species in maintaining balance, providing insights into sustainable agricultural practices. The ODE systems is:

Figure 15: Population Dynamics with Bats $A(t)$

$$\frac{dP}{dt} = \left(\frac{r_p + \frac{\delta_p A}{1+kA}}{1+C_p} \right) P \left(1 - \frac{P + \gamma Z + wH}{K_P} \right),$$

$$\frac{dZ}{dt} = r_z Z \left(1 - \frac{Z + \alpha P}{K_Z} \right),$$

$$\frac{dH}{dt} = r_h H \left(1 - \frac{H + \beta B}{K_H} \right) - C_p H - \delta_h A H,$$

$$\frac{dB}{dt} = r_b B \left(1 - \frac{B}{K_B} \right),$$

$$\frac{dA}{dt} = r_a A \left(1 - \frac{A}{K_A} \right) + \delta_a P A - \mu_a A.$$

As Figure 15 shows, crops (P) initially experience rapid growth, stabilizing at a high population size (~ 110) due to the pollination benefits provided by bats and the suppression of pests. Meanwhile, weeds (Z) rise moderately as they compete with crops for resources but eventually stabilize at a lower level (~ 60) due to the dominance of crops. Pests (H) are suppressed by the combined predation from bats (A) and beneficial insects (B), with their population declining and stabilizing around 15. Beneficial insects (B) stabilize around 30 after an initial adjustment, playing a steady role in pest control. Bats (A) exhibit slower growth compared to other species but stabilize below 20, effectively balancing their roles as insectivores and pollinators. The confidence intervals represent the variability in population sizes, highlighting the sequence of adjustments as the ecosystem achieves stability, transitioning to a state of reduced chemical dependency and enhanced ecological resilience.

4.3 Multi-Objective Optimizations for Agricultural Ecosystem Management

Organic farming emphasizes sustainable practices that eliminate synthetic chemicals, fertilizers, and genetically modified organisms. Key techniques such as crop rotation, cover cropping, and composting enhance soil fertility and structure. Integrated Pest Management (IPM) relies

on natural pest control, while biological methods reduce the need for chemical pesticides. To further optimize pest management, we can model the use of pesticide by defining decision variables that represent pesticide inputs.

The decision variables represent the quarterly pesticide input over 4 years:

$$\mathbf{x} = [x_1, x_2, \dots, x_{16}],$$

where $x_i \in [0, 3]$, which is the pesticide input in the i -th quarter.

The pesticide concentration at time t is given by our previous assumptions:

$$C(t) = \sum_{i=1}^n x_i C_0 e^{-k(t-iT)}, \quad t \geq iT,$$

where C_0 is the initial concentration of pesticide, k is the decay rate of the pesticide, and T is the time interval per quarter.

The population dynamics of crops (P), pests (H), beneficial insects (B), and soil vitality (N) are governed by the **tritrophic and soil vitality dynamics model** introduced in section 2.1.3.

4.3.1 Objective Function 1: Economic objective function

The economic objective function is defined as:

$$F_1(\mathbf{x}) = \text{Harvest Revenue} - \text{Pesticide Cost},$$

where **Harvest Revenue** is given by $\text{Harvest Revenue} = \sum_{i=0}^3 0.86 \cdot P(4i) \cdot \text{Crop Price}$, representing the income generated from harvested crops, and **Pesticide Cost** is given by $\text{Pesticide Cost} = \sum_{i=1}^{16} x_i \cdot \text{Pesticide Cost/Unit}$, which accounts for the total expenses associated with pesticide use.

Harvest Disturbance

At the end of each year (quarter $4i$, $i = 1, 2, 3, 4$), the crop population is reduced by the harvest:

$$P(t_{\text{harvest}}^+) = P(t_{\text{harvest}}^-) - \text{Harvest}_i,$$

where:

$$\text{Harvest}_i = 0.86 \cdot P(4i).$$

The data used in this analysis is sourced from the USDA Crop Production report titled *TotalAcresU.S.Planted/Harvested – June28*. This report provides detailed statistics on the area planted and harvested for various crops across the United States for the years 2023 and 2024. Using this data, we calculated the harvest ratio for each crop, defined as the ratio of harvested area to planted area.

We analyzed the data in Table 1 to calculate the average harvest ratios for 2023 and 2024, representing the overall efficiency of crop production for each year. To provide a comprehensive measure of harvesting efficiency over the two-year period, we determined the overall average harvest ratio by taking the mean of the yearly averages. The calculated harvest ratio for 2023 is approximately 0.855, and for 2024 is 0.865. Based on these results, we selected **0.86** as a representative value for the harvest ratio.

Crop	Planted 2023	Harvested 2023	Ratio 2023	Planted 2024	Harvested 2024	Ratio 2024
Barley	3101	2555	0.824	2557	2017	0.789
Corn for Grain	94641	86513	0.914	91475	83438	0.912
Oats	2555	831	0.325	2299	872	0.379
Rice	2894	2854	0.986	2943	2900	0.985
Rye	2293	322	0.141	2204	378	0.172
Sorghum for Grain	7195	6115	0.850	6405	5390	0.841
Wheat, All	49575	37272	0.752	47240	38785	0.821
Winter Wheat	36699	24683	0.672	33805	25808	0.764
Durum Wheat	1676	1604	0.957	2165	2092	0.966
Other Spring Wheat	11200	10985	0.981	11270	10885	0.966
Canola	2344.5	2319.2	0.989	2662.5	2616	0.982
Flaxseed	178	160	0.899	140	125	0.893
Mustard Seed	245	238.1	0.972	218	203.5	0.933
Peanuts	1645	1574	0.957	1757	1702	0.969
Safflower	129.5	126	0.973	127	117	0.921
Soybeans for Beans	83600	82356	0.985	86100	85261	0.990
Sunflower	1315	1267.5	0.964	898.5	862.6	0.960
Dry Edible Beans	1180	1156.9	0.980	1359	1318	0.970
Dry Edible Peas	966	941	0.974	1033	988	0.957
Lentils	546	523	0.958	836	790	0.945

Table 1: Crop Harvest Ratios

4.3.2 Objective Function 2: Environmental objective function

The environmental objective function combines soil fertility and biodiversity:

$$F_2(\mathbf{x}) = \frac{w_1}{16} \sum_{n=1}^{16} S_n + \frac{w_2}{16} \sum_{n=1}^{16} H_n,$$

The **Soil Fertility Index** (SFI) [2] is a measure of soil health and productivity, influenced by chemical inputs (C) and soil properties (S), such as pH, cation exchange capacity (CEC), and nitrogen content (N). The index in our model is defined as:

$$S_n = \text{SFI}(C, S) = \frac{\text{SFI}_{\max}}{1 + \alpha R_n \cdot (1 - \beta \cdot g(S))}, \quad R_n = \sum_{i=1}^n x_i C_0 e^{-k(n-iT)},$$

where $\text{SFI}_{\max}$ is the maximum potential soil fertility value, α is a sensitivity parameter for chemical input, n represents the degree of nonlinearity ($n > 1$), and β is an adjustment coefficient for soil property effects ($0 < \beta < 1$). The soil property impact function $g(S)$ (range 0-1, closer to 1 indicates greater soil property deviation) quantifies deviations from optimal conditions and is given by:

$$g(S) = w_1 \cdot \frac{(pH_{\text{opt}} - pH)^2}{pH_{\text{opt}}^2} + w_2 \cdot \frac{(\text{CEC}_{\text{opt}} - \text{CEC})^2}{\text{CEC}_{\text{opt}}^2} + w_3 \cdot \frac{(N_{\text{opt}} - N)^2}{N_{\text{opt}}^2}$$

where w_1 , w_2 , and w_3 are weights for pH, CEC, and nitrogen content, respectively. The Soil Fertility Index integrates the effects of chemical inputs and soil properties, providing a comprehensive measure of soil health. High chemical inputs (C) can reduce fertility, especially when soil properties deviate from optimal conditions, as quantified by $g(S)$. The index highlights the trade-offs between chemical use and soil quality, emphasizing the importance of maintaining balanced soil properties for sustainable agricultural productivity.

The **Shannon Index** [6], a measure of biodiversity, quantifies the diversity of species within an ecosystem by considering both species richness and evenness. It is defined as:

$$H_n = - \sum_i p_i \ln(p_i), \quad p_P = \frac{P}{P + H + B}, \quad p_H = \frac{H}{P + H + B}, \quad p_B = \frac{B}{P + H + B},$$

where p_P , p_H , and p_B represent the proportions of crops (P), pests (H), and beneficial insects (B) relative to the total population. A higher Shannon Index indicates greater biodiversity, reflecting a balanced community with multiple species and no single dominant group, while a lower value suggests reduced diversity, often due to the dominance of one or a few species. In the context of the model, the Shannon Index is used to evaluate the biodiversity of crops, pests, and beneficial insects, providing insights into the ecosystem's stability and health, particularly in scenarios such as the removal of herbicides. This index helps assess the impact of agricultural practices on ecosystem balance and species diversity.

4.3.3 Normalized objective functions

To normalize $F_1(\mathbf{x})$, S_n and H_n , we scale their values to the range $[0, 1]$. This ensures consistency in the evaluation of soil fertility and biodiversity, facilitating their integration into the environmental objective function. Below are the normalization formulas and their explanations:

The economic objective $F_1(\mathbf{x})$ and other indices are normalized using a min-max approach to ensure comparability across objectives. The normalized economic objective is defined as:

$$\tilde{F}_1(\mathbf{x}) = \frac{F_1(\mathbf{x}) - F_{1,\min}}{F_{1,\max} - F_{1,\min}},$$

where $F_{1,\min}$ and $F_{1,\max}$ are the minimum and maximum expected values of $F_1(\mathbf{x})$, respectively, ensuring $\tilde{F}_1(\mathbf{x}) \in [0, 1]$.

Soil fertility S_n is normalized as:

$$\tilde{S}_n = \frac{S_n - \min(S)}{\max(S) - \min(S)},$$

where $\min(S)$ and $\max(S)$ are the minimum and maximum values of soil fertility, respectively, ensuring $\tilde{S}_n \in [0, 1]$.

The normalized Shannon biodiversity index is given by:

$$\tilde{H}_n = \frac{H_n}{\ln(3)},$$

where H_n is the current index, and $\ln(3)$ is its theoretical maximum for three species (crops, pests, and beneficial insects). This ensures $\tilde{H}_n \in [0, 1]$, making it comparable to other normalized indices.

Using the normalized values $\tilde{S}_n$ and $\tilde{H}_n$, the environmental objective function becomes:

$$\tilde{F}_2(\mathbf{x}) = \frac{w_1}{16} \sum_{n=1}^{16} \tilde{S}_n + \frac{w_2}{16} \sum_{n=1}^{16} \tilde{H}_n,$$

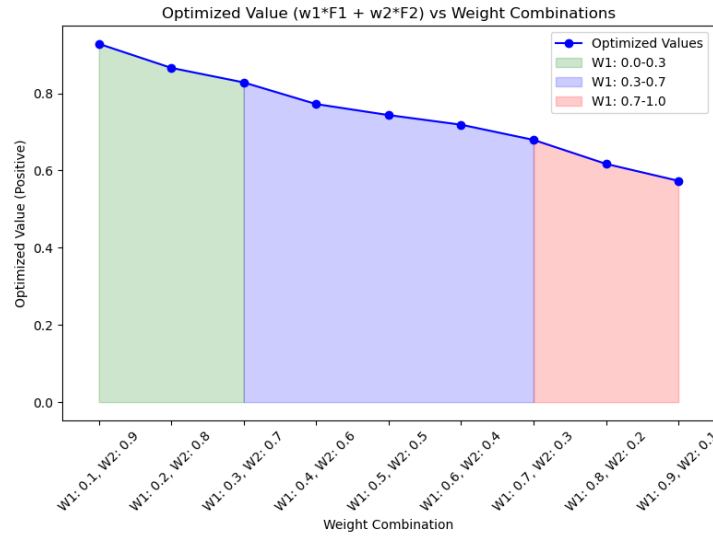


Figure 16: Optimization of $\omega_1 \tilde{F}_1(\mathbf{x}) + \omega_2 \tilde{F}_2(\mathbf{x})$ vs Weight Combination

4.3.4 Two cases of optimization

The multi-objective optimization problems are defined as two cases:

$$\text{Maximize } \omega_1 \tilde{F}_1(\mathbf{x}) + \omega_2 \tilde{F}_2(\mathbf{x}),$$

and

$$\text{Maximize } \tilde{\mathbf{F}}(\mathbf{x}) = [\tilde{F}_1(\mathbf{x}), \tilde{F}_2(\mathbf{x})],$$

subject to:

$$x_i \in [0, 3], \quad \forall i \in \{1, 2, \dots, 16\}, \quad \omega_1 + \omega_2 = 1.$$

In the first case of our optimization problem, we tested multiple computational methods, including SLSQP, L-BFGS-B, COBYLA, and Nelder-Mead—to identify the most effective strategy for balancing economic and ecological objectives in agricultural management. Through systematic comparison, SLSQP (Sequential Least Squares Programming) emerged as the optimal method, demonstrating superior performance in convergence speed, solution accuracy, and handling complex constraints.

The analysis of the optimization model, supported by data from Figures 16 and 17, reveals that ecological benefits, such as soil fertility and biodiversity, play a more critical role in achieving sustainable agricultural outcomes compared to economic gains. Figure 16 shows that as the weight for ecological objectives (w_2) increases, the optimized value improves significantly, with the optimized value rising from approximately 0.58 to 0.90 as w_2 shifts from 0.1 to 0.9. This indicates that ecological benefits have a stronger influence on the system's long-term sustainability.

Furthermore, Figure 17 illustrates the comparison between initial and optimized pesticide application strategies, demonstrating that the optimized profiles are smoother and more evenly distributed over time, reducing fluctuations in pesticide use. For example, in Guess 1, the initial pesticide input varies erratically, while the optimized strategy stabilizes pesticide use at a consistent level, reducing unnecessary fluctuations and minimizing environmental harm. Similarly, in Guess 2, the optimized strategy smooths out an initially increasing trend, ensuring

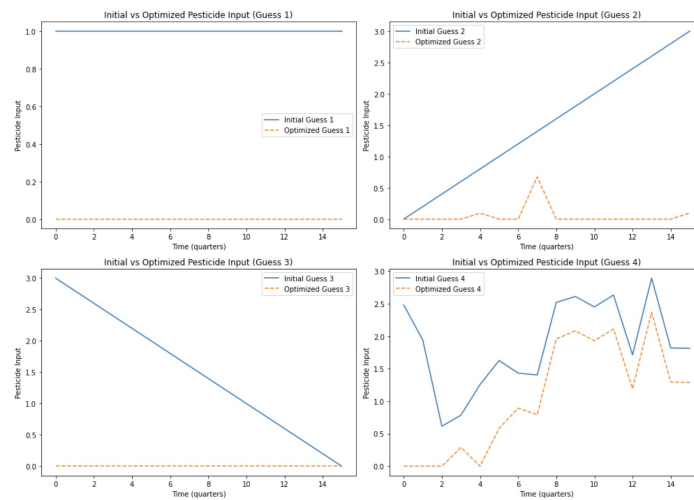


Figure 17: Initial vs Optimized Pesticide Input

a more balanced and sustainable approach. These results underscore that dynamic adjustments in pesticide application, guided by the optimization model, not only maintain crop yield but also significantly reduce ecological damage. Therefore, the data unequivocally supports the need to prioritize ecological objectives in agricultural practices, as this approach ensures a harmonious balance between economic productivity and environmental health, paving the way for long-term sustainability and resilience in agricultural systems.

In the second case of our optimization problem, we employ **NSGA-II** (Non-dominated Sorting Genetic Algorithm II) [3] to address a multi-objective optimization problem where we aim to optimize two conflicting objectives: economic benefits and ecological sustainability. The economic objective considers maximizing revenue from crop harvests while minimizing pesticide costs, whereas the ecological objective focuses on improving soil fertility and biodiversity. These two objectives are inherently at odds—actions that maximize one often lead to trade-offs with the other. For instance, higher pesticide use may increase short-term crop revenue but can

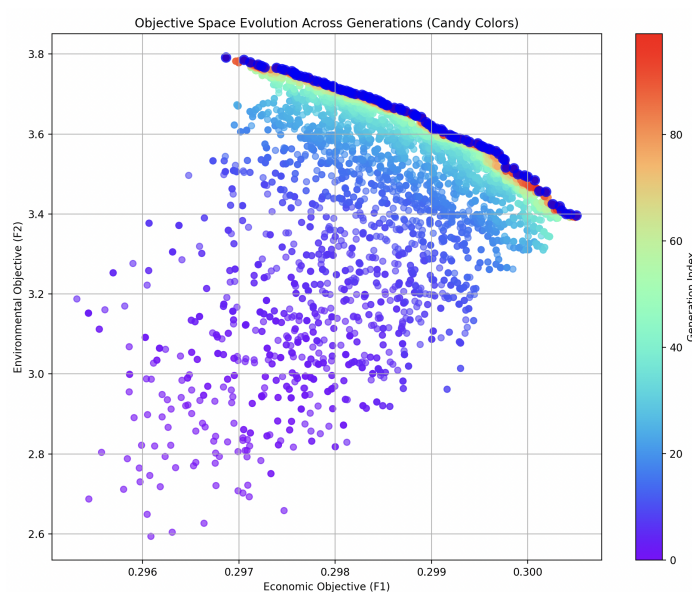


Figure 18: Pareto Frontier of Ecological and Economic Benefits

harm biodiversity and degrade soil quality, reducing ecological benefits.

To resolve this conflict, we frame it as a multi-objective optimization problem, where the goal is not to simultaneously maximize both objectives but to identify a set of solutions that represents the best possible trade-offs. We utilize NSGA-II to solve this multi-objective optimization problem. NSGA-II employs non-dominated sorting to classify solutions based on their performance across both objectives. A solution is said to *dominate* another if it performs better in at least one objective without being worse in the other. The algorithm also uses genetic operators, such as crossover and mutation, to explore the solution space efficiently, and crowding distance is applied to maintain diversity in the solution set.

The output of this optimization process is a **Pareto front**, which is a set of non-dominated solutions. Each solution on the Pareto front represents a trade-off where improving one objective would necessarily degrade the other. For example, one solution on the Pareto front might favor higher crop revenue with moderate ecological impacts, while another might prioritize ecological benefits at the expense of some economic gains. These solutions dominate all others in the sense that no alternative solution exists that performs better in both objectives simultaneously.

The Pareto front thus provides decision-makers with a spectrum of optimal trade-offs, shown as Figure 18. Farmers can choose a solution from the Pareto front based on their specific priorities—whether they prioritize short-term economic returns, long-term ecological health, or a balanced approach. Figure 19 showcases some of the optimal solutions for chemical usage over four years by two different kinds of plots, including line plot and bar chart. This approach ensures that no better solution exists outside the Pareto front, offering a comprehensive understanding of the trade-offs involved in sustainable agricultural management.

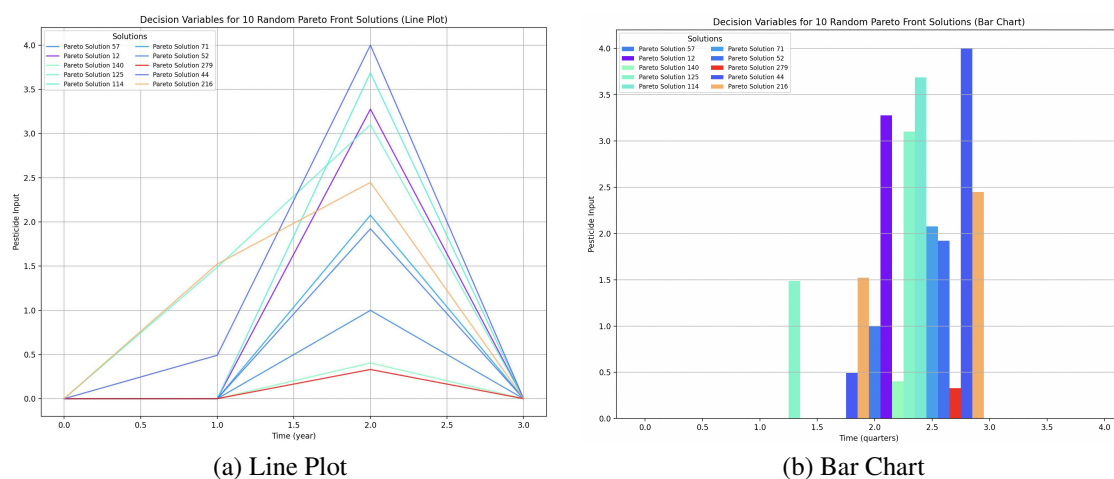


Figure 19: Decision Variables for 10 Random Pareto Front Solutions

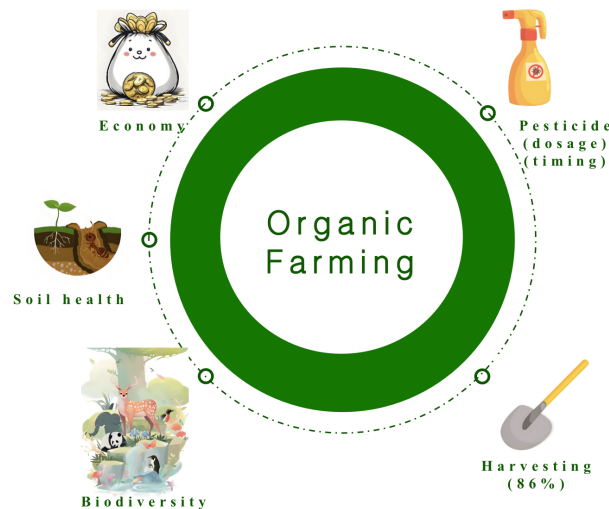


Figure 20: Organic Farming

5 One-Page Letter

Dear Farmer,

We hope this letter finds you well. As you are exploring organic farming practices, we recommend you for taking a significant step toward sustainable agriculture. Recently, we conducted a detailed analysis regarding both natural processes and human decisions for agricultural management, and we also explored the multi-objective optimization models for a better way to balance between economic success and environmental goals. We'd like to share some insights which might support you in adopting practical, sustainable approaches for your farm.

A key aspect of sustainable farming is the dynamic management of both pesticide use and harvest schedules. Appropriate dosage and timing of pesticide helps minimize chemical inputs while maintaining their effectiveness, thus generating greatest economic value. Besides, it also helps avoid the significant fluctuations often seen in conventional farming, which can degrade soil quality and harm biodiversity. Similarly, optimizing harvest schedules is vital for preventing the over-exploitation of soil nutrients. Research shows that maintaining an optimal harvest ratio of around 85% leaves sufficient crop residue in the field, providing essential soil cover to prevent erosion and support long-term soil health.

Another essential aspect of organic farming is enhancing soil health, such as pH balance, nitrogen content, and cation exchange capacity. This can be achieved through methods like composting, incorporating organic matter, and practicing crop rotation to replenish vital nutrients. Reducing dependence on chemical fertilizers further helps preserve soil fertility. By prioritizing soil health, you can not only boost current productivity but also ensure the land remains fertile and viable for future generations.

Biodiversity is another vital component of building a resilient agricultural system. Our analysis shows that ecosystems with greater species evenness are more stable and resilient. For example, introducing bats can effectively control pest populations, such as mosquitoes and moths, while also enhancing pollination, significantly boosting crop yields. By harnessing these

natural ecological services, you can reduce reliance on chemical interventions while creating a healthier and more productive farming environment.

As shown in Figure 20, we summarized five essential elements in organic farming: economy, soil health, pesticide use, harvesting, and biodiversity. While the transition to organic farming may result in higher upfront costs and potential short-term revenue loss compared to conventional methods, the long-term benefits outweigh these trade-offs. Prioritizing ecological objectives such as soil fertility, biodiversity, and reduced pesticide use will lead to lower input costs over time. Additionally, the resilience of healthy soil allows crops to better withstand climate variability, leading to more stable yields. Furthermore, organic products often command higher market prices, improving your profitability in the long run.

In conclusion, by implementing these strategies, you can effectively balance economic productivity with environmental sustainability. If you would like further details or assistance in implementing these strategies, please don't hesitate to reach out. I wish you the best of success in your journey toward sustainable farming.

Best regards,

Member / Consideration of Mature Agricultural Practice

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A Report on Use of AI

1. Open AI ChatGPT (Jan 25, 2025 Version, ChatGPT-4)

Question: Please introduce the concept of organic farming.

Answer:

Organic farming is a sustainable agricultural practice that relies on natural processes, biodiversity, and local ecological balance instead of synthetic inputs like chemical fertilizers, pesticides, and genetically modified organisms (GMOs). It emphasizes soil health, natural pest control, diverse crop systems, and animal welfare while minimizing environmental impact. By avoiding synthetic chemicals, organic farming produces healthier food and supports ecological resilience, but it faces challenges such as lower yields, higher costs, and certification barriers. Despite these challenges, its focus on sustainability and growing consumer demand makes it an essential approach to environmentally friendly and health-conscious food production.

2. Open AI ChatGPT (Jan 26, 2025 Version, ChatGPT-4)

Question: Please introduce the Soil Fertility Index (SFI) and Shannon Index

Answer:

The **Soil Fertility Index (SFI)** is a quantitative metric used to assess the fertility status of soil by integrating various physical, chemical, and biological properties that influence plant growth and productivity. It helps farmers, researchers, and policymakers understand the overall health and nutrient availability of soil, aiding in sustainable land management and agricultural planning.

Key components of the SFI include:

- **Soil Nutrients:** Levels of essential nutrients like nitrogen (N), phosphorus (P), potassium (K), and micronutrients.
- **Soil Organic Matter:** The presence of organic materials that contribute to soil fertility and microbial activity.
- **Soil pH:** The acidity or alkalinity, which affects nutrient availability.
- **Physical Properties:** Parameters like soil texture, moisture, and bulk density.
- **Biological Indicators:** Microbial activity, enzyme activity, and earthworm populations.

The SFI provides a composite score that reflects the soil's capacity to support sustainable crop production and ecosystem functions.

The **Shannon Index**, also known as the Shannon-Weiner or Shannon Diversity Index, is a mathematical measure of biodiversity that reflects both the richness and evenness of species in a community. It is widely used in ecology and environmental science to assess the diversity of plant, animal, or microbial populations.

The formula for the Shannon Index is:

$$H = - \sum_{i=1}^S p_i \ln(p_i)$$

where:

- S : Total number of species (species richness).
- p_i : Proportion of individuals or abundance of the i -th species relative to the total number of individuals in the community.

Key features of the Shannon Index:

- **Higher Values:** Indicate greater biodiversity, with species more evenly distributed.
- **Lower Values:** Indicate lower biodiversity or dominance of a few species.

The Shannon Index is useful in monitoring environmental changes, evaluating habitat health, and comparing biodiversity across ecosystems. In agriculture, it can help measure the diversity of crops, soil microbes, or pest populations, linking biodiversity to ecosystem resilience and productivity.

3. Open AI ChatGPT (Jan 27, 2025 Version, ChatGPT-4)

Question: What are the applications of NSGA-II method?

Answer:

NSGA-II (Non-dominated Sorting Genetic Algorithm II) is one of the most widely used multi-objective optimization algorithms. It aims to optimize multiple conflicting objectives simultaneously, producing a set of solutions called the **Pareto front**. NSGA-II is favored for its efficiency, simplicity, and ability to maintain a diverse set of solutions.

1. **Multi-objective Optimization:** Unlike single-objective optimization, multi-objective optimization involves optimizing two or more conflicting objectives simultaneously (e.g., maximizing profit while minimizing cost). The solution is a set of trade-offs where no single solution is strictly better than the others in all objectives, known as the *Pareto optimal front*.
2. **Non-dominated Sorting:** Solutions are ranked based on the concept of domination:
 - A solution A dominates B if A is better than or equal to B in all objectives and strictly better in at least one objective.

Solutions are sorted into hierarchical levels:

- **Front 1:** Solutions that are not dominated by any others.
- **Front 2:** Solutions that are only dominated by those in Front 1, and so on.

3. **Crowding Distance:** Measures the diversity of solutions in a front. For each solution, the crowding distance is calculated based on its neighbors in the objective space. Solutions in less crowded regions are preferred to ensure a diverse Pareto front.
4. **Genetic Algorithm Framework:** NSGA-II uses the principles of evolutionary algorithms, including selection, crossover, and mutation, to evolve the population toward the Pareto-optimal front.

NSGA-II has been successfully applied in various fields, including:

- **Engineering Design:** Optimizing designs with multiple conflicting objectives, such as weight and strength.
- **Finance:** Portfolio optimization considering risk and return.
- **Energy Systems:** Balancing energy efficiency, cost, and environmental impact.
- **Logistics:** Solving multi-objective vehicle routing and supply chain problems.
- **Bioinformatics:** Optimizing biological network models and genetic studies.